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GENERATIVE AI AND AI PLATFORM ENGINEERING (25CS2207E)

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ABSTRACT

Section / Team Details

Section No.: 8

Team No.: 23

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Project Title: BERT vs GPT vs T5 – Model Selection for HR Automation

Team Members

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1. Abstract

The project "BERT vs GPT vs T5 – Model Selection for HR Automation" aims to identify the suitability of different Natural Language Processing (NLP) models for automating selected Human Resource (HR) activities. HR departments perform tasks such as screening resumes, identifying suitable candidates, responding to employee queries, and preparing summaries of HR-related documents. Performing these activities manually can require considerable time and effort.

This project studies and compares three transformer-based models: BERT, GPT, and T5, by assigning each model to an appropriate HR task. BERT is used for resume text classification and candidate-job matching, GPT is used to generate responses to common employee HR queries, and T5 is used to summarize resumes and other HR documents. Each workflow is evaluated using suitable task-specific measures, along with processing time and resource efficiency.

The project aims to analyse the strengths and limitations of these models and determine which is most suitable for each selected HR activity, rather than assuming that one

model is best for every task. The proposed system is intended to reduce repetitive manual work, support faster HR operations, and improve consistency while keeping important decisions under human review. Privacy, fairness, and responsible use of candidate and employee information are also considered.

2. Problem Statement

Traditional HR activities such as resume screening, candidate shortlisting, answering employee queries, and reviewing HR documents often involve repetitive manual effort. These processes can be time-consuming and may lead to delays or inconsistent handling of information.

Although models such as BERT, GPT, and T5 offer different language-processing capabilities, their strengths are not identical. A model that performs well in text classification may not be the best choice for generating conversational answers or summarizing documents.

Therefore, the project focuses on implementing and evaluating BERT for resume classification and candidate matching, GPT for HR question answering, and T5 for document summarization. The models will be assessed using appropriate task-specific evaluation measures, processing time, and efficiency to understand their suitability for the selected HR activities.

3. Objectives

- To understand the architecture and capabilities of BERT, GPT, and T5.
- To implement the three models for selected HR automation tasks.
- To classify resume information and support candidate-job matching using BERT.
- To build an HR question-answering workflow using GPT.
- To summarize resumes and HR documents using T5.
- To evaluate each model using suitable task-specific measures, processing time, and efficiency.
- To identify the strengths, limitations, and suitable use cases of each model.
- To consider privacy, fairness, and human review in the proposed HR automation workflow.

4. Software Components

Software Component	Purpose
Python	Main programming language
Google Colab / VS Code	Development environment

BERT Model	Resume classification and candidate matching
GPT Model	HR chatbot and response generation
T5 Model	Resume summarization and text generation
Hugging Face Transformers	Loading and implementing AI models
Pandas	Dataset processing and analysis
Scikit-learn	Model evaluation and performance comparison
Streamlit	Developing the user interface
Evaluation Metrics	Task-specific model evaluation and comparison
Google Colab / VS Code	Development environment

5. Hardware Components

Hardware Component	Purpose
Computer / Laptop	Running the project
Processor (Intel i3/i5 or equivalent)	Executing program operations
RAM (Minimum 8 GB)	Handling datasets and model processing
Hard Disk / SSD (256 GB or above)	Storing datasets and project files
GPU (Optional)	Faster AI model processing
Internet Connection	Accessing pre-trained models and datasets
Keyboard and Mouse	User interaction

6. Expected Outcome

The expected outcome is a prototype HR automation system that demonstrates three task-based workflows using BERT, GPT, and T5 and presents a comparative analysis of their suitability for the selected HR activities.

The system is expected to:

- Classify resume information and support candidate-job matching using BERT.
- Generate relevant responses to common employee HR queries using GPT.
- Produce concise summaries of resumes and HR documents using T5.
- Evaluate each workflow using suitable task-specific measures, such as precision, recall, and F1-score for classification; answer correctness and relevance for question answering; and summary quality measures for summarization.
- Compare processing time and resource requirements under the tested setup.
- Present the strengths, limitations, and most suitable use case of each model.

The final output will be a working prototype and a comparative report that helps explain which model is more appropriate for each selected HR task. The system is intended to assist HR personnel; final recruitment and employee-related decisions will remain subject to human review.